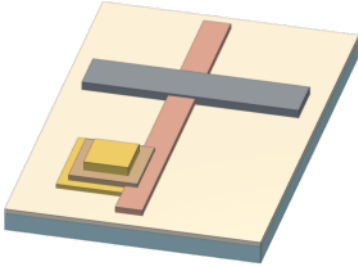


Memory structures: planar cells to vertical arrays

DSL-authored schematic geometry; dimensions chosen for legibility

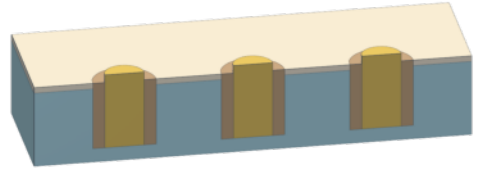
(a) Planar 1T1C DRAM



patterned word/bit lines + stacked capacitor

Dennard, US 3,387,286 (1968)

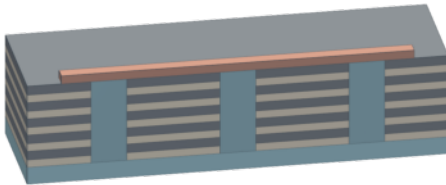
(b) Deep-trench DRAM



circular deep etch + exact dielectric/electrode fills

Mandelman et al., IBM JRD 46 (2002)

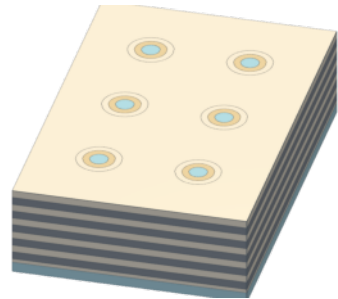
(c) Vertical-BL 3D DRAM



repeated stack + circular depth etch + Si:P fill

Oh et al., IEEE VLSI (2024)

(d) BiCS 3D NAND



word-line stack + nested charge-trap fills

Tanaka et al., IEEE VLSI (2007)